MORTALITY RATES& DEATH CERTIFICATE

MORTALITY RATES

- It la possible to be obtained and calculated since there is a system for registration of deaths, this deaths are affected by occurrence of a disease, so it can be used as incidence of Health of the community.

Crude Death Rate

$$= \frac{\text{No. of Deaths during the year}}{\text{Mid year population}} \times 1000$$

Age specific death rate (ASDR):

= Total no of deaths of a certain age group in certain year and certain placeX 1000

Mid year population of the same age group in the same year and place

Infant mortality Rate (IMR) =

$$\frac{\text{No. of Infant (less than 1 year) deaths during the year}}{\text{No. of live births during the year}} \times 1000$$

Neo- Natal Mortality Rate (NM Rate)

$$= \frac{\text{No. of infant deaths less than 29 days during the year}}{\text{No. of live births during the year}} \times 1000$$

Post Natal mortality Rate (PNM Rate)

$$= \frac{\text{No. of deaths of 29 days - 1 year}}{\text{No. of live births during the year}} \times 1000$$

Children 1- 4 years mortality Rate

$$= \frac{\text{No. of deaths of children aged 1-4 years in a given locality and year}}{\text{No. of children of the same age group, locality and year}} \times 1000$$

CAUSES OF INFANT AND CHILD MORTALITY:

1- Acute respiratory infections (ARI)

2- Diarrhea

3- Congenital anomalies

4- Low birth weight (LBW)

5- Accidents

Still Birth rate (SBR) =

$$\frac{\text{No. of Still Births during the year}}{\text{No. of live births + No. of still births during the year}} \times 1000$$

Peri- Natal mortality Rate (PM Rate)

$$= \frac{\text{No. of Still Births \& Infant death of less than 7 day}}{\text{No. of live births + No. of still births during the year}} \times 1000$$

4. Maternal Mortality Rate =

$$\frac{\text{No. of Maternal deaths during the year}}{\text{Number of females aged (15-49 years) in one year}} \times 100{,}000$$

Maternal Mortality ratio =

$$\frac{\text{No. of Maternal deaths during the year}}{\text{No. of live births during the year}} \times 100{,}000$$

CAUSES OF MATERNAL MORTALITY

- **Direct Obstetric Causes of Death:**
  - **1- Hemorrhage (69%): Antepartum, Postpartum or Trauma**
  - **2. Genital Sepsis (19%)**
  - **3. Hypertensive Diseases of Pregnancy without or with convulsion (8%)**
  - **4. Abortion (1.5%)**
  - **5. Others (2.5%): Ectopic pregnancy, Rupture uterus, Obstructed labor, Caesarian section, Anaesthesia or Pulmonary embolism.**

- **B. Indirect Obstetric Causes of Death:**

1. Anemia.
2. Cardiovascular diseases.
3. Neurological disorder (e.g. epilepsy).
4. Infections and parasitic diseases (e.g. Bilharziasis)
5. Digestive disorders.
6. Urogenital as pregnancy may aggravate the renal problem.
7. Neoplasms.
8. Others (e.g. respiratory system diseases).

Case Fatality Rate (CFR):

$$= \frac{\text{All deaths assigned to a specific disease}}{\text{All cases of that disease in the same period and area}} \times 1000$$

- **It predicts the risk of dying if the disease is contracted.**

Cause Specific Mortality Rate (CSMR):

$$= \frac{\text{All deaths assigned to a specific disease}}{\text{Estimated mid-interval population of the same locality}} \times 1000$$

Proportionate Mortality Ratio (PMR):

$$= \frac{\text{All deaths assigned to a specific disease}}{\text{Total deaths from all causes in the same time period}} \times 1000$$

- **PMR usually expressed as a percentage.**
- **it is the percent proportion of the number of deaths of a particular disease to total deaths in a certain locality (or country, or group of the population) and year.**
- **Proportionate death ratio is valuable to find out the relative mortality role of each disease, for a particular group (e.g. infants and preschool children) or the population, arrange causes of death by magnitude, and so find the leading causes of death.**

Population	All Ages	Under 5 years	5- < 15	Females 15 - <25	Femal es 25- <35	Female s 35- <45	45+
Midyear Population	1000,000	50,000	55,000	200,000	150,00 0	100,000	50,000
Live Births	15000						
Deaths	10000	700	400	300			
No of infant deaths less than 29 days	50						
No of infant deaths 29 days to one year	450						
Infant deaths during the first year	500						
No of still births	300						
Maternal deaths during the year	5500						

- **Calculate:**
  - **Crude death rate.**
  - **Infant mortality rate.**
  - **Neonatal mortality rate.**
  - **Maternal mortality ratio.**
  - **Age specific death rate 5-15 years.**
  - **Post-neonatal mortality rate.**

Crude death rate.

$$= \frac{\text{No. of Deaths during the year}}{\text{Mid year population}} \times 1000$$

$$= \frac{10000}{1000000} \times 1000 = 10 \text{ per } 1000$$

Infant mortality rate.

$$= \frac{\text{No. of Infant deaths during the year}}{\text{No. of live births during the year}} \times 1000$$

$$= \frac{500}{15000} \times 1000 = 33.33 \text{ per } 1000$$

Neonatal mortality rate.

$$= \frac{\text{No. of infant deaths less than 29 days during the year}}{\text{No. of live births during the year}} \times 1000$$

$$= \frac{50}{15000} \times 1000 = 3.33 \text{ per } 1000$$

- **Maternal mortality ratio.**

$$= \frac{\text{No. of Maternal deaths during the year}}{\text{No. of live births during the year}} \times 100{,}000$$

$$= \frac{5500}{15000} \times 100{,}000 = 36.7 \text{ per } 100{,}000$$

Age specific death rate 5-15 years.

$$= \frac{\text{No. of Deaths in a particular age group}}{\text{Mid year population of same age group}} \times 1000$$

$$= \frac{400}{55000} \times 1000 = 7.3 \text{ Per } 1000$$

- **Post-neonatal mortality rate.**

$$= \frac{\text{No. of deaths of 29 days - 1 year}}{\text{No. of live births during the year}} \times 1000$$

$$= \frac{450}{15000} \times 1000 = 30 \text{ per } 1000$$

DEATH CERTIFICATION

- **An International Form of Medical Certification of Death has been developed and is in practice in most countries of the world**

THE DEATH CERTIFICATE IS DIVIDED INTO THREE SECTIONS:

- 1. Part I—including diseases or conditions directly leading to death and antecedent causes
- 2. Part II—other significant conditions
- 3. a column to record the approximate interval between onset and death.

Causes of death		Approximate Interval between onset and death
I. Disease or condition directly leading to death*	a) ..... (Due to or as a consequence of)	.....
Antecedent causes Morbid conditions, if any, giving rise to the above cause, stating the underlying condition last	b) ..... Due to or as a consequence of	.....
	c) ..... Due to or as a consequence of	.....
	d) ..... Due to or as a consequence of	.....
II. Other significant conditions contributing to death, but not related to diseases or conditions causing it	....	....
*This does not mean the mode of dying, e.g., heart failure, respiratory failure. It means the disease, injury or complication that caused death		

IN PART I:-

- The cause leading to death stated on line (a),
- Leading (direct) casuse of death is disease or condition directly leading to death "leading cause of death"
- the antecedent conditions (reported on line (b), (c) & (d) that gave rise to the cause reported in line (a) i.e the underlying cause being stated last in the sequence of events.
- No entry is necessary in lines (b) and (c) if the disease or condition directly leading to death, stated in line (a), describes completely the chain of events.

UNDERLYING CAUSE OF DEATH IS

- the disease or injury which initiated the train of morbid events leading directly to death; or
- the circumstances of the accident or violence which produced the fatal injury.

IN PART II

- Any other significant condition that unfavorably influenced the course of the morbid process and thus contributed to fatal outcome, but that was not related to the disease or condition directly causing death.

Bleeding oesophageal varices ← Immediate cause of death

▲

Portal hypertension

▲

Liver cirrhosis

▲

Hepatitis B ← Underlying cause of death

Cerebral haemorrhage ←	Immediate cause of death
Secondary hypertension	
Chronic pyelonephritis due to outflow obstruction	
Prostatic adenoma ←	Underlying cause of death
Diabetes mellitus ←	Contributory condition

CASE STUDY

- A 60-year-old hypertensive patient was admitted to the surgical casualty ward with severe abdominal pain and vomiting. She was diagnosed as having strangulated femoral hernia with a bowel perforation. She underwent surgery to release the hernia and resect the intestine, with an end-to-end anastomosis. Two days after the surgery she developed signs of peritonitis and she died 2 days later.

Causes of death		Approximate Interval between onset and death
I. Disease or condition directly leading to death*	a) ..... (Due to or as a consequence of)	.....
Antecedent causes Morbid conditions, if any, giving rise to the above cause, stating the underlying condition last	b) ..... Due to or as a consequence of	.....
	c) ..... Due to or as a consequence of	.....
	d) ..... Due to or as a consequence of	.....
II. Other significant conditions contributing to death, but not related to diseases or conditions causing it	....	....
*This does not mean the mode of dying, e.g., heart failure, respiratory failure. It means the disease, injury or complication that caused death		

INTERNATIONAL FORM OF MEDICAL CERTIFICATE OF CAUSE OF DEATH

	Cause of death	Approximate interval between onset and death
I Disease or condition directly leading to death*	(a) Peritonitis	2 days
	due to (or as a consequence of)	
Antecedent causes Morbid conditions, if any, giving rise to the above cause, stating the underlying condition last	(b) Strangulated femoral hernia with bowel perforation	2 weeks
	due to (or as a consequence of)	
	(c)	
	due to (or as a consequence of)	
	(d)	
II Other significant conditions contributing to the death, but not related to the disease or condition causing it	Hypertension	unknown

** This does not mean the mode of dying, e.g. heart failure, respiratory failure. It means the disease, injury or complication that caused death.*

Thank you